Calibration Pilot 31 August 2026

Sample. 126 completers (NOVICE +2: 35, NOVICE +3: 32, EXPERT +2: 33, EXPERT +3: 26); n per block within participant, NOVICE $\in \{3, 4, 5\}$, EXPERT $\in \{15, 20, 30\}$, eight rounds each.

- **The stimulus works.** Bets track the Bayesian posterior in every growth arm (Figure 1), yet people still land convinced on the wrong side of the truth on 20.5% of rounds, and 16.6% on the flat Stable islands.
- **Pick growth $\beta = +2$ coconuts and novice $n = 3$ dots.** The +2 arm manufactures the most false signal (exposure 69.5% vs 49.0%, accuracy at chance 48.5%); the catch is it fools experts nearly as much as novices, so +3 is the role-separation arm. Novice $n = 3$ is the most seductive small sample: a flat island looks significantly sloped 29.5% of the time at $n = 3$ against 13.9% at $n = 5$.
- **Pick n=20 dots. Expert n does not change how much doubt experts state, only how often their confidence is misplaced.** The persuadable middle is flat across expert n (25.4%/23.1%/24.8% at $n = 15/20/30$); what moves is how often confidence is misplaced (convinced-wrong 21.4% at $n = 15$ to 15.3% at $n = 30$) and how often the evidence itself is genuinely open (posterior band 11.0% to 7.2%). Both sides in §3.
- **The persuadable middle exists in both roles.** Experts and novices sit unsure (bet $\in (30, 70)$) on about a quarter of rounds (24.4% of expert, 26.3% of novice); in a two-novice group a confidently-wrong voice meets a movable listener 11.8% of the time (§2).
- **Attrition is concentrated at the quiz.** 5 left during the instructions, 27 abandoned at the quiz, and 23 were disqualified at the quiz.
- **Timings.** Median 6.3 min from the first instructions page to passing the comprehension quiz, and a median of 8.4 s per task round.

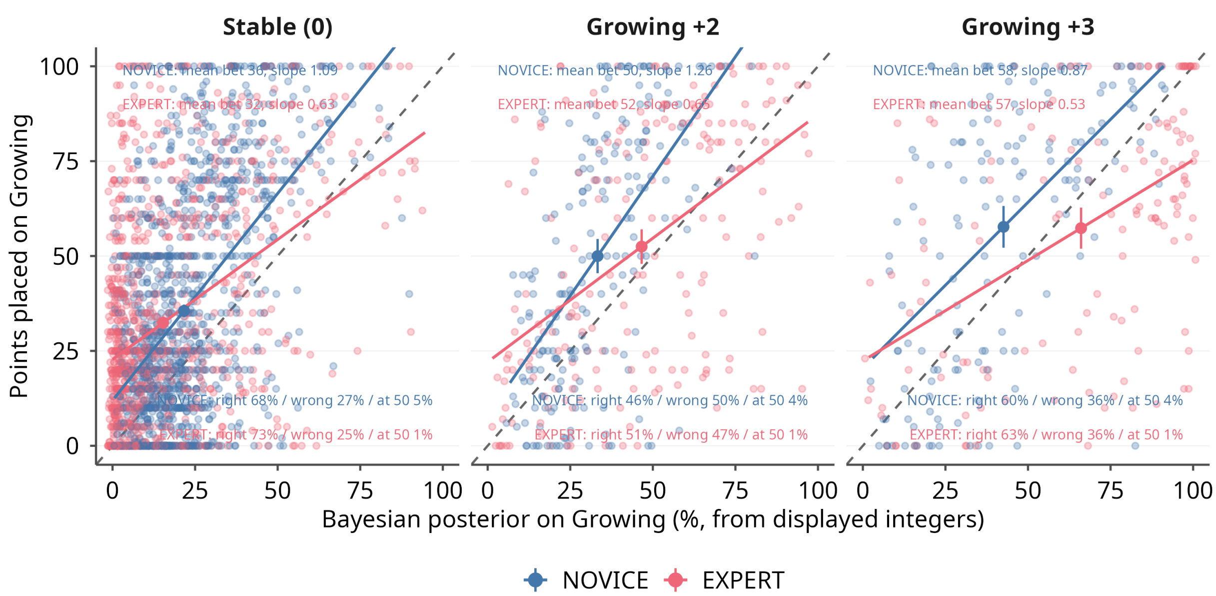


Figure 1: Points placed on Growing against the Bayesian posterior, split by realised growth arm.

1 Q1: Is the stimulus effective?

Two definitions, used throughout:

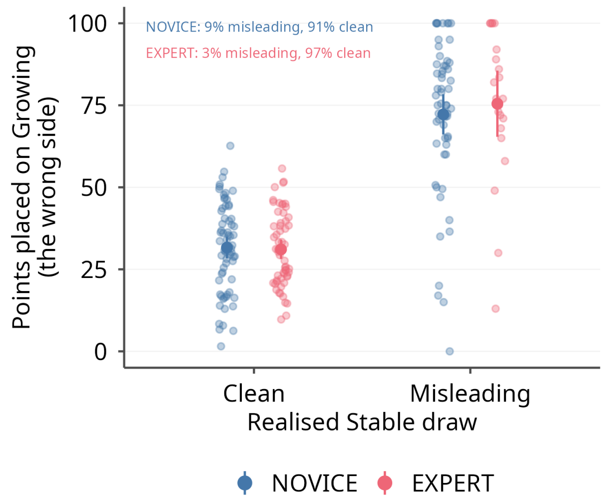
- **The three-way stated split** partitions every round (overall 54.1% / 25.4% / 20.5%, summing to 100):
  - **convinced-right** (70+ points on the true side);
  - **persuadable** (bet strictly between 30 and 70, i.e. within 20 points of an even 50/50, ties at 50 counted here);
  - **convinced of the wrong side** (70+ points on the wrong side).

  The persuadable middle is the movable mass a discussion could shift, measured in §2.
- **Misleading draw:** a true-Stable (flat) stimulus whose fitted slope is nonetheless statistically significant ($|t| > 2$); a **clean** draw is one that is not. This is the same false-signal definition used in the n tables in §3.
- **Two benchmarks for "tricked"** (Table 1): wrong versus the **true state** (the headline) and wrong against the participant's own **Bayesian posterior** (the irrationality read). The meeting picks which one the study should target.
  - **They diverge on Growing.** On 60.9% of Growing rounds the posterior itself favours Stable (weak arms plus the strong 0.75-Stable prior), so wrong-vs-truth on Growing (43.3%) is mostly rational prior-following; wrong against the posterior is only 7.1%.
  - **They nearly coincide on Stable,** so a convinced-wrong Stable bet is a genuine mistake: convinced-wrong runs 16.6% there.
- **Role split (the study's premise):** novices over-react and experts under-react to the realised evidence (reaction slopes 1.460 vs 0.665, Bayes = 1), and experts read the flat island a little more accurately (73.4% vs 68.1% correct, p=.058).
- **Confidence on the wrong side reflects differences in exposure to misleading draws, not susceptibility to a misleading draw** (Fig. 2). A misleading flat draw pulls both roles onto Growing by about the same amount (NOVICE +40.2 vs EXPERT +42.9 points, panel a); small novice samples simply produce far more of them, because the few-dot novice regime concentrates at higher linear-fit quality (panel c) and so shows a significant slope by chance more often: the misleading-draw share is 17.2% at novice $n = 3$ against 5.9% at expert $n = 20$.
  - **The Growing side mirrors it** (panel b): growth stays hidden ($|t| \leq 2$) on 72.5% of novice and 60.9% of expert Growing rounds, and on those hidden draws both roles hedge near 50 (mean bet 45.5 novice, 44.3 expert) rather than following the Stable-heavy prior down, the Growing-side face of base-rate neglect.

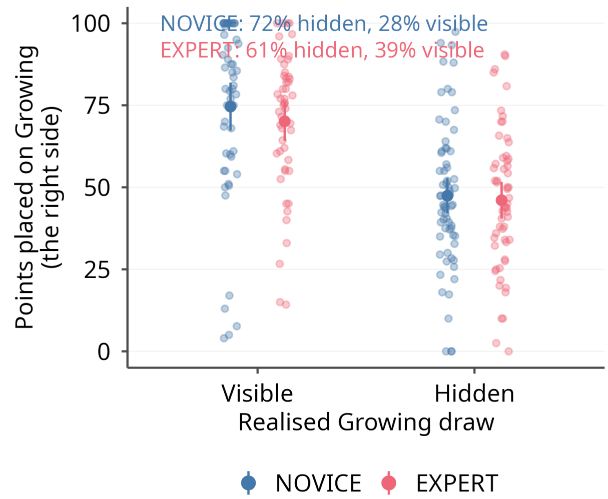
Table 1: How often the stimulus misleads, by role and true state.

Role	State	Rounds	Maj. wrong (%)	Conv. wrong (%)	Maj. vs post. (%)	Conv. vs post. (%)
NOVICE	Stable	1250	27.3	18.4	23.0	14.3
NOVICE	Growing	358	43.9	33.2	1.1	0.8
EXPERT	Stable	1068	25.3	14.4	21.1	11.2
EXPERT	Growing	348	42.8	33.6	13.2	9.8
All	all	3024	30.3	20.5	18.6	11.1

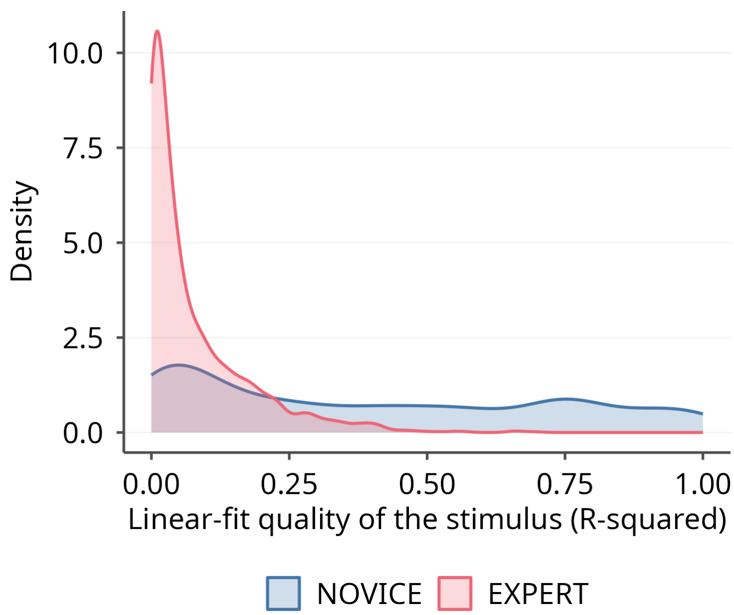
Notes: Convinced = 70+ points on the wrong side. The "vs post." columns count only rounds where the posterior itself pointed to the true state, so the evidence did not excuse the error.



(a) Stable: bet on Growing, clean vs misleading draws



(b) Growing: bet on Growing, visible vs hidden draws



(c) linear-fit quality (R^2)

Figure 2: The mirrored seduction of flat and growing draws, and the fit quality each role meets.

Notes: Points placed on Growing, by role. A *misleading* Stable draw has a significant positive fitted slope ($|t| > 2$); a *hidden* Growing draw has no significant slope ($|t| \leq 2$), growth that the displayed integers do not reveal. Panel (a): mean bet on clean versus misleading Stable draws. Panel (b): mean bet on visible versus hidden Growing draws; the printed shares give each role's exposure, the fraction of its true-Growing rounds on which growth stayed hidden. Panel (c): overlaid kernel densities of the R^2 of the line fitted to the displayed integers; the few-dot novice regime concentrates higher, so a novice sample carries a spurious slope, flat or growing, by chance more often.

1.1 What predicts conviction

- Linear-fit quality and the wrong-side fraction move bet extremity beyond the posterior; raw slope strength does not (Table 2).

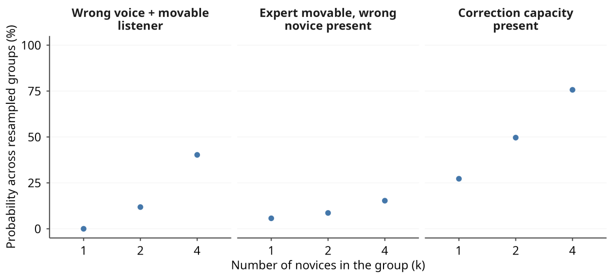
Table 2: Rank correlation of bet extremity with each stimulus feature.

Stimulus feature	Rounds	Spearman rho	p
Linear-fit quality (R-squared)	3024	0.067	p<.001
Slope t-ratio (strength over noise)	2973	0.066	p<.001
Slope magnitude	3024	0.026	p=.151
Slope standard error	3024	-0.105	p<.001
Wrong-side fraction (messiness)	3009	-0.062	p<.001

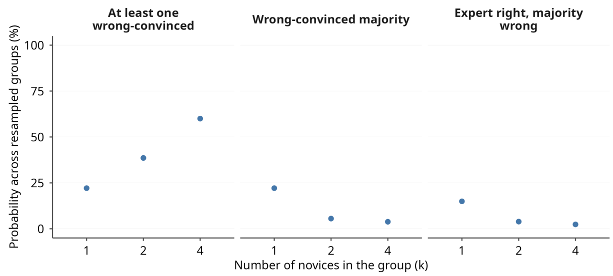
Notes: Bet extremity is $|a - 50|$. p two-sided.

2 Interesting cases and group composition

A round is **interesting** when a novice lands convinced-wrong.



(a) persuadability events



(b) group composition

Figure 3: A wrong voice and a movable listener in the same resampled group, by group size.

- **Reading the stated split.** A stated bet partitions each round into *convinced-right* (70+ points on the true side), *persuadable* (bet in $(30, 70)$) and *convinced-wrong* (70+ on the wrong side), so the three shares sum to 100. Confidence is summarised by the median bet extremity $|a - 50|$ and the prior-anchored mean |logit deviation|; the rational benchmark *Post. .40–.60* is the share of rounds whose Bayesian posterior sits within 0.10 of the fence (Table 3).
- **The panel events.** Panel (a): *wrong voice + movable listener* is ≥ 1 convinced-wrong novice together with ≥ 1 persuadable novice; *expert movable, wrong novice present* is a persuadable expert with ≥ 1 convinced-wrong novice; *correction capacity present* is ≥ 1 convinced-right member beside ≥ 1 persuadable member. Panel (b) reads the presence, majority and expert-right-while-novices-wrong configurations by k.
- **The persuadable middle is real and role-symmetric** (Table 3): overall 54.1% convinced-right, 25.4% persuadable, 20.5% convinced-wrong; the persuadable share is 26.3% for novices and 24.4% for experts (tighter 40–60 band, 12.0% and 9.5%).
- **Convinced-wrong seeds are plentiful in a group** (Fig. 3b): a novice is convinced-wrong on 21.7% of the 1608 novice rounds, so at $k = 4$ a group holds one 60.0% of the time (against 22.1% for a lone novice). A convinced-wrong *majority* stays rare and falls with k (3.8% at $k = 4$); an expert right while the novices are mostly wrong occurs 14.9% of the time at $k = 1$.
- **Movable listeners are plentiful once the group is more than a pair** (Fig. 3a): a wrong voice meets a persuadable listener 40.2% of the time at $k = 4$ (against 0.0% at $k = 1$), a group of four holds 1.0 persuadable novices on average, and a convinced-right member sits beside a persuadable one 75.7% of the time.
- **The expert-gets-dragged case is rarer** (Fig. 3a): the expert is himself persuadable with a convinced-wrong novice present 15.3% of the time at $k = 4$ (from 5.7% at $k = 1$). Since the expert's stated persuadable middle does not shrink with n, this exposure is set by the arm choice only weakly.

Table 3: Convinced-right / persuadable / convinced-wrong shares, and confidence, by role and n.

Role	Dots n	Rounds	Conv. right (%)	Persuadable (%)	Conv. wrong (%)	Med. |a-50|	Mean |logit dev|
NOVICE	3	536	51.9	25.9	22.2	30.0	2.030
NOVICE	4	536	52.2	26.3	21.5	30.0	1.962
NOVICE	5	536	51.8	26.7	21.5	29.0	1.779
EXPERT	15	472	53.2	25.4	21.4	30.0	1.891
EXPERT	20	472	56.1	23.1	20.8	30.0	1.645
EXPERT	30	472	59.9	24.8	15.3	30.0	1.798

Notes: Completer rounds; novice $n \in \{3, 4, 5\}$ then expert $n \in \{15, 20, 30\}$. The three stated-split shares partition each row and sum to 100. Column definitions are in the section bullets.

3 Q2: Which parameters?

- **Growth arm: +2, and the catch is it fools experts nearly as much as novices: do we want experts to be reliably correct or not?** (Table 4). +2 maximises the false-confidence raw material (false-signal exposure 69.5% vs 49.0%, accuracy at chance 48.5% vs 61.5%). If the design needs experts to be a reliable signal, +3 is the role-separation arm.

Table 4: Growth-arm decision facts (Growing rounds).

Growth arm	Rounds	Novice conv-wrong (%)	Expert conv-wrong (%)	False signal (%)	On true side (%)
Growing +2	410	35.1	37.1	69.5	48.5
Growing +3	296	30.7	28.7	49.0	61.5

Notes: Convinced-wrong = 70+ points on the wrong side. "False signal" is the share of Growing rounds whose posterior still favours Stable. The two convinced-wrong columns show experts fooled almost as often as novices on +2.

- **Novice** $n = 3$ (Table 5). The lever is the design false-signal curve, a function of n alone: a flat island looks significantly sloped 29.5% of the time at $n = 3$ against 13.9% at $n = 5$, so $n = 3$ is the most seductive small sample.
- **Expert** $n = 20$**: expert** n **changes reliability, not stated doubt.** The persuadable middle is flat across n (25.4%/23.1%/24.8%, median $|a-50|$ 30.0/30.0/30.0); what moves is the wrong tail (convinced-wrong 21.4% to 15.3%) and the rational posterior band (11.0% to 7.2% near the fence), not stated openness (Table 3).
- **Reliability side:** $n = 30$ drives expert convinced-wrong to 15.3% (from 21.4% at $n = 15$), false signal to the floor (3.7% of Stable rounds), and accuracy up to 75.0%. An expert who then under-reacts is identifiably mistaken, not reading a deceptive sample.
- **Convincibility side:** $n = 15/20$ leaves experts convinced-wrong 21.4%/20.8% of the time and more exposed to false signal (6.2%/5.9% vs 3.7% at $n = 30$), so more experts hold a stated *wrong* position to be argued out of.

Table 5: Convinced-wrong, false-signal on Stable and accuracy, by role and n.

Role	Dots n	Rounds	Conv-wrong (%)	False signal, Stable (%)	On true side (%)
NOVICE	3	536	22.2	17.2	64.4
NOVICE	4	536	21.5	17.7	63.8
NOVICE	5	536	21.5	16.1	64.7
EXPERT	15	472	21.4	6.2	64.4
EXPERT	20	472	20.8	5.9	67.8
EXPERT	30	472	15.3	3.7	75.0

Notes: Completer rounds; novice $n \in \{3, 4, 5\}$ then expert $n \in \{15, 20, 30\}$. Convinced-wrong = 70+ points on the wrong side. "False signal, Stable" is the share of flat-island rounds that look significantly sloped ($|t| > 2$). The realised column is nearly flat within each regime; the wave is too small to resolve the curve, though it confirms the direction.